

Sarah Abe

Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with 8 years of experience designing and managing the full data lifecycle, from initial requirements to final deployment in production environments.

Expert in Cloud Platforms, including AWS, Azure, and GCP, with a proven track record of building scalable data pipelines and cost-effective cloud infrastructure.

Specialist in Big Data technologies, using Spark, Hive, and Scala to process massive datasets and transform raw data into actionable business insights.

Highly proficient in Python and SQL, utilizing libraries like PySpark, Boto3, and Pandas to automate complex ETL processes and improve data accuracy.

Advanced Data Architect skilled in designing Relational and Dimensional models, including Star and Snowflake schemas, to support high-performance analytics.

Strong background in NoSQL and Linux, with hands-on experience managing Cassandra and HBase databases within secure, containerized environments.

Collaborative Technical Leader focused on writing clean, maintainable code and building reliable data solutions that drive informed business decisions.

WORK EXPERIENCE

Infosys – New York, NY

Senior Data Engineer (Azure Data Platform)

Jan 2022 – Present

At Infosys, I work on building and modernizing enterprise data platforms using Microsoft Azure technologies. My focus is on scalable ETL pipelines, data modeling, and analytics solutions.

Azure Data Lake and ETL Modernization

- Designed and developed ETL pipelines using **Azure Data Factory** to load data from SFTP, REST APIs, SQL databases, and flat files into **Azure Data Lake Storage**.
- Built transformation logic using **Azure Databricks (PySpark and Spark SQL)**.
- Implemented **Delta Lake** for reliable and version-controlled data processing.

- **Modernized enterprise data architecture** by migrating legacy on-premise systems to Azure, reducing operational infrastructure costs by 20% within the first year
- Monitored and maintained daily pipeline runs and resolved production issues.
- Optimized large-scale data processing in Azure Databricks using PySpark, which improved query performance by 40% and significantly reduced daily processing windows and Implemented Delta Lake to establish a reliable "Source of Truth," enabling version control and ACID transactions for critical financial reporting data.

Enterprise Data Warehouse on Azure

- **Mentored a team of 5 junior engineers** on Databricks best practices and Spark optimization, increasing overall team productivity by 15%.
- Designed dimensional data models using **Star Schema and Snowflake Schema** and built curated data layers to support reporting and analytics teams.
- Created dashboards and reports using **Power BI** and Installed and configured **Azure SQL Database and Synapse Analytics**.
- Automated deployments using **Azure DevOps and CI/CD pipelines** allowing the team to deploy updates 3x faster with fewer production errors.

Workflow Automation and Data Governance

- Designed and maintained **Apache Airflow jobs** to manage workflow dependencies and developed data validation scripts using Python and SQL.
- Documented data flows, architecture, and transformation logic.
- Guided junior engineers on Databricks best practices and certification preparation.

Environment: Azure Data Factory, Azure Databricks, Delta Lake, Azure Data Lake, Azure SQL, Synapse, Airflow, Python, PySpark, SQL, Power BI

Walmart Inc. – Bentonville, AR

Data Engineer (AWS Platform)

Jun 2019 – Dec 2021

At Walmart, I supported large-scale retail data systems using AWS and Big Data technologies. I worked mainly on building reliable batch and streaming pipelines.

Real-Time Data Processing

- Enabled AWS Managed Streaming for Kafka (MSK) for real-time ingestion, allowing for instant inventory updates across store locations.
- Built Python-based consumers to process streaming data and scheduled workflows using Apache Airflow.
- Replaced legacy SSIS workflows with modern Spark-based solutions on EMR, increasing system flexibility.

AWS Data Lake Implementation

- Built end-to-end data pipelines using **AWS S3, AWS Glue, and EMR** and Processed large datasets using **Apache Spark and Hive**.
- Developed ETL jobs in PySpark to transform raw data into curated datasets.
- Managed access control and security for S3 buckets and tuned Spark jobs to reduce processing time and improve efficiency.

Reporting and Analytics Support

- Built dimensional models for business reporting and developed SQL queries for analytics and operational dashboards.
- Created dashboards using **Tableau and AWS QuickSight** and replaced legacy SSIS workflows with Spark-based solutions on EMR.
- Designed and maintained **high-performance dimensional models** (Star and Snowflake schemas) in Amazon Redshift to support large-scale business reporting.
- Developed complex **SQL queries and stored procedures** to generate real-time operational dashboards, improving data accessibility for stakeholders.
- Created interactive data visualizations using **Tableau and AWS QuickSight**, allowing executive leadership to track store performance and inventory trends.
- Successfully **replaced legacy SSIS workflows** with modern Spark-based solutions on EMR, reducing daily data processing time by over 30%.
- **Environment:** AWS (S3, EMR, Glue, Redshift, MSK, EC2), Spark, Hive, Airflow, Python, SQL, Tableau, QuickSight

Cognizant Technology Solutions – Dallas, TX

Big Data Developer

Jan 2018 – May 2019

At Cognizant, I worked on Hadoop systems to help the company store and move large amounts of data. My daily job involved setting up ways to bring data in from old databases, making sure the data was clean, and helping the team start moving their work over to AWS cloud servers.

- **Moved large amounts of data** from Oracle and MySQL databases into Hadoop (HDFS) using a tool called **Sqoop**.
- **Built and managed Hive tables** and wrote SQL-style queries to help the business teams get the reports they needed.
- **Made data jobs run faster** by using partitioning tricks in MapReduce, which saved time during daily processing.
- **Checked Hadoop logs every day** to fix any errors and made sure the system stayed up and running for everyone.
- **Designed data workflows in Informatica BDM** to keep track of changing data (SCD Type 1 and Type 2) in our tables.

- **Set up automatic checks** to compare source data with target data so we knew the information was correct and nothing was missing.
- **Wrote simple Linux scripts** to keep an eye on file uploads from SFTP folders and send alerts if something failed.
- **Helped the team move data to the cloud** by setting up folders in **AWS S3** and running Spark jobs on **AWS EMR**.
- **Set up and fixed Linux-based EC2 servers** in AWS to host our data applications.
- **Used Git to save my code** and worked with the team in **JIRA** to stay on track with our weekly tasks.

Environment: Hadoop, Hive, Sqoop, MapReduce, Informatica, AWS (S3, EMR, EC2), Linux, Python, SQL, Git, JIRA.

CERTIFICATIONS

- Microsoft Certified: Azure Data Engineer Associate (DP-203)
- AWS Certified Data Engineer – Associate
- Databricks Certified Data Engineer Professional
- Snowflake Core Certified SnowPro Core

TECHNICAL SKILLS

- **Cloud:** Azure (ADF, Databricks, Synapse), AWS (S3, EMR, Glue, Redshift), GCP.
- **Languages:** Python (PySpark, Pandas), SQL, Scala, Shell Scripting.
- **Tools:** Apache Spark, Hive, Airflow, Delta Lake, Kafka, Sqoop, Informatica.
- **Databases:** Snowflake, Azure SQL, PostgreSQL, Cassandra, HBase, MongoDB.

EDUCATION

- **Bachelor of Science in Computer Science**
- **City University of New York (CUNY) – Brooklyn College** | *Graduated May 2017*